



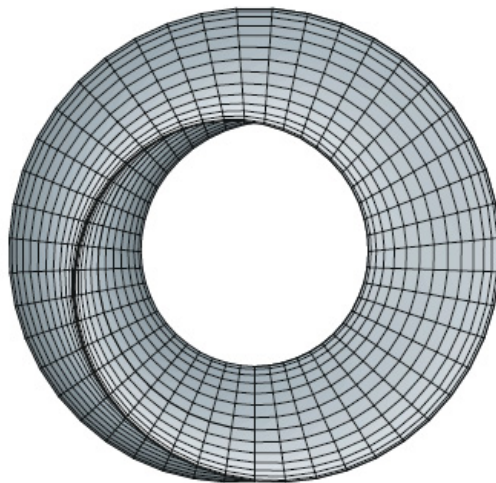
UNIVERSITY OF ZIMBABWE

HMTHCS111 CALCULUS 2

Calculus of Several Variables

Author:
G. NHAWU

Department:
MATHEMATICS AND
COMPUTATIONAL SCIENCES



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Chapter 7

Multiple Integration

There was a young man from Trinity, Who solved the square root of infinity. While counting the digits, He was seized by the fidgets, Dropped science, and took up divinity.

—Author Unknown

7.1 Iterated Integrals

In the previous chapter, we saw that it is meaningful to differentiate functions of several variables with respect to one variable while holding the other variable constant, we can integrate functions of several variables by a similar procedure. For example, if we have the partial derivative $f_x(x, y) = 2xy$, then by considering y constant, we can integrate with respect to x to obtain

$$\begin{aligned} f(x, y) &= \int f_x(x, y) dx && \text{Integrate with respect to } x \\ &= \int 2xy dx && y \text{ is held constant} \\ &= y \int 2x dx && \text{Factor out constant } y \\ &= y(x^2) + C(y) && \text{Anti-derivative of } 2x \text{ is } x^2 \\ &= x^2 + C(y). && C(y) \text{ is function of } y \end{aligned}$$

Note that the constant of integration, $C(y)$ is a function of y . In other words, by integrating with respect to x , we are able to recover $f(x, y)$ only partially. For example, by considering y constant, we can apply the Fundamental Theorem of calculus to evaluate

$$\int_1^{2y} 2xy dx = x^2 y \Big|_1^{2y} = (2y)^2 y - (1)^2 y = 4y^3 - y.$$

Note that the variable of integration cannot appear in either limit of integration. For example, it makes no sense to write $\int_0^x y \, dx$.

Example 7.1.1. Evaluate $\int_1^x (2x^2y^{-2} + 2y) \, dy$.

Solution: Considering x to be constant and integrating with respect to y produces

$$\begin{aligned} \int_1^x (2x^2y^{-2} + 2y) \, dy &= \left[\frac{-2x^2}{y} + y^2 \right]_1^x \\ &= \left(\frac{-2x^2}{x} + x^2 \right) - \left(\frac{-2x^2}{1} + 1 \right) \\ &= 3x^2 - 2x - 1. \end{aligned}$$

Notice that in the above example the integral defines a function of x and can *itself* be integrated.

Example 7.1.2. Evaluate $\int_1^2 \left[\int_1^x (2x^2y^{-2} + 2y) \, dy \right] dx$.

Solution:

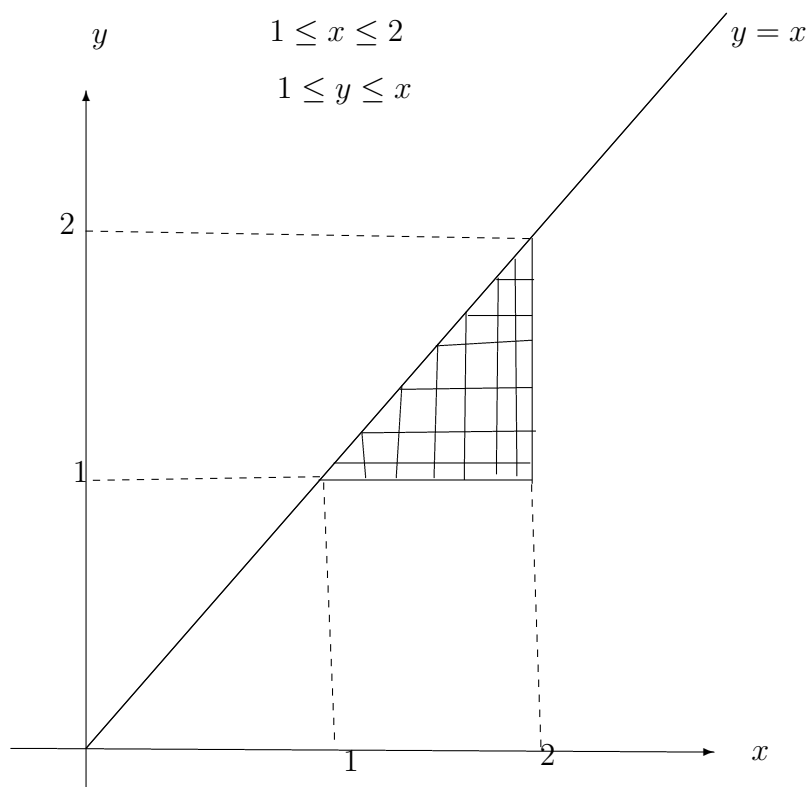


Figure 7.1: The region of integration

Using the result from the previous example, we have

$$\begin{aligned}\int_1^2 \left[\int_1^x (2x^2y^{-2} + 2y) dy \right] dx &= \int_1^2 (3x^2 - 2x - 1) dx \\ &= \left[x^3 - x^2 - x \right]_1^2 \\ &= 2 - (-1) \\ &= 3.\end{aligned}$$

The integral in the above example is an **iterated integral**. Iterated integrals are usually written simply as

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx \quad \text{and} \quad \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

The **inside limits of integration** can be variable with respect to the outer variable of integration. However, the **outside limits of integration** must be constant with respect to both variables of integration. After performing the inside integration, we obtain a definite integral, and the second integration produces a real number.

One order of integration will often produce a simpler integration problem than the other order. The order of integration affects the ease of integration, but not the value of the integral.

7.1.1 Comparing Different Orders of Integration

Example 7.1.3. Sketch the region whose area is represented by the integral

$$\int_0^2 \int_{y^2}^4 dx dy.$$

Then find another iterated integral using the order $dy dx$ to represent the area, and show that both integrals yield the same value.

Solution: From the given limits of integration, we know that

$$y^2 \leq x \leq 4 \quad \text{Inner limits of integration}$$

which means that the region R is bounded on the left by the parabola $x = y^2$ and on the right by the line $x = 4$. Furthermore, because

$$0 \leq y \leq 2 \quad \text{Outer limits of integration}$$

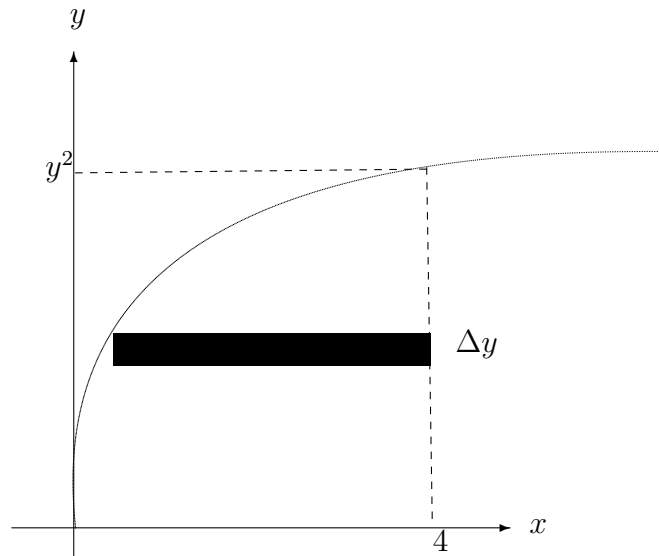


Figure 7.2: Fig 1

we know that R bounded below by the x -axis. The value of this integral is

$$\begin{aligned} \int_0^2 \int_{y^2}^4 dx dy &= \int_0^2 x \Big|_{y^2}^4 dy \\ &= \int_0^2 (4 - y^2) dy \\ &= \left[4y - \frac{y^3}{3} \right]_0^2 = \frac{16}{3}. \end{aligned}$$

To change the order of integration to $dydx$, place a vertical rectangle in the region. From this we can see that the constant bounds $0 \leq x \leq 4$ serve as the outer limits of integration. By solving for y in the equation $x = y^2$, we can conclude that the inner bounds are $0 \leq y \leq \sqrt{x}$. Therefore, the area of the region can be represented by

$$\int_0^4 \int_0^{\sqrt{x}} dy dx.$$

By evaluating this integral, we can see that it has the same value as the original integral.

$$\begin{aligned} \int_0^4 \int_0^{\sqrt{x}} dy dx &= \int_0^4 y \Big|_0^{\sqrt{x}} dx \\ &= \int_0^4 \sqrt{x} dx \\ &= \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^4 = \frac{16}{3}. \end{aligned}$$

Example 7.1.4. Express $\int_0^4 \int_{\frac{x}{2}}^2 e^{y^2} dy dx$ as an iterated integral with order of integration reversed and evaluate.

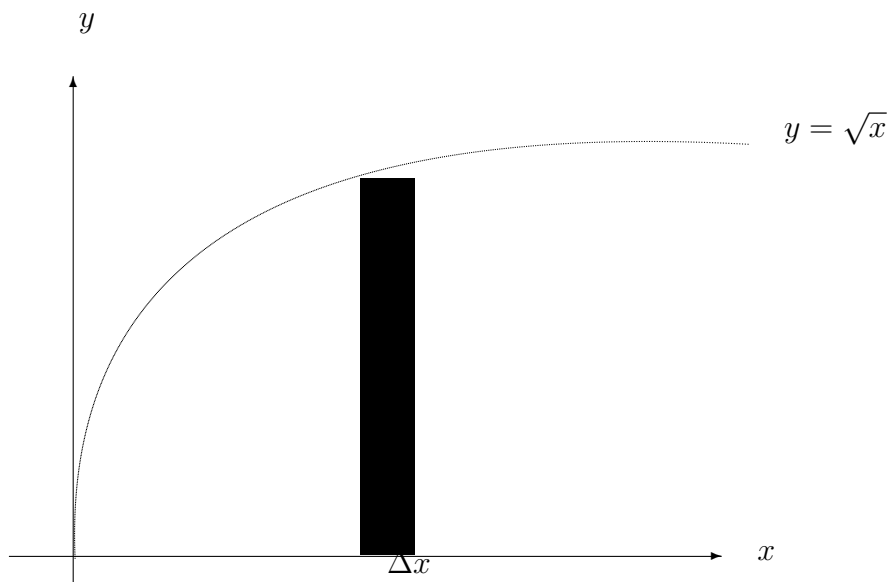


Figure 7.3: Fig 2

Solution: From the given limits of integration we see that, for a fixed x , y varies from $y = \frac{x}{2}$ to $y = 2$ and x varies from $x = 0$ to $x = 4$. We can also describe the region as, for y fixed, x varies from $x = 0$ to $x = 2y$ and y varies from $y = 0$ to $y = 2$. The corresponding iterated integral is $\int_0^2 \int_0^{2y} e^{y^2} dx dy$. Solving, we have

$$\begin{aligned} \int_0^2 \int_0^{2y} e^{y^2} dx dy &= \int_0^2 x e^{y^2} \Big|_{x=0}^{x=2y} dy \\ &= \int_0^2 2y e^{y^2} dy \\ &= e^{y^2} \Big|_0^2 \\ &= e^4 - 1. \end{aligned}$$

Example 7.1.5. Evaluate $\int_0^2 \int_{\frac{y}{2}}^1 y e^{x^3} dx dy$.

Solution: We cannot integrate first with respect to x , as indicated, because it happens that e^{x^3} has no elementary anti-derivative. So we try to evaluate the integral by first reversing the order of

integration.

$$\begin{aligned}
 \int_0^2 \int_{\frac{y}{2}}^1 ye^{x^3} dx dy &= \int_0^1 \int_0^{2x} ye^{x^3} dy dx \\
 &= \int_0^1 \left[\frac{1}{2} y^2 \right]_0^{2x} x e^{x^3} dx \\
 &= \int_0^1 2x^2 e^{x^3} dx \\
 &= \left[\frac{2}{3} e^{x^3} \right]_0^1 \\
 &= \frac{2}{3}(e - 1).
 \end{aligned}$$

7.2 Double Integrals and Volume

7.2.1 Double Integrals and Volume of a Solid Region

Definition of Double Integral

If f is defined on a closed, bounded region R in the xy -plane, then the **double integral of f over R** is given by

$$\iint_R f(x, y) dA = \lim_{|\Delta| \rightarrow 0} \sum_{i=1}^n f(x_i, y_i) \Delta x_i \Delta y_i$$

provided the limit exists. If the limit exists, then f is **integrable** over R .

A double integral can be used to find the volume of a solid region that lies between the xy -plane and the surface given by $z = f(x, y)$.

Volume of a Solid Region

If f is integrable over a plane region R and $f(x, y) \geq 0$ for all (x, y) in R , then the volume of the solid region that lies above R and below the graph of f is defined as

$$V = \iint_R f(x, y) dA.$$

Example 7.2.1. Find the volume of the solid region R bounded by the surface

$$f(x, y) = e^{-x^2}$$

and the planes $y = 0$, $y = x$ and $x = 1$.

Solution: The base of R in the xy -plane is bounded by the lines $y = 0$, $x = 1$ and $y = x$. The two possible orders of integration are

$$\int_0^1 \int_0^x e^{-x^2} dy dx \quad \text{and} \quad \int_0^1 \int_y^1 e^{-x^2} dx dy.$$

By setting up the corresponding iterated integrals, we can see that the order $dx dy$ requires the anti-derivative $\int e^{-x^2} dx$, which is not an elementary function. On the other hand, the order $dy dx$ produces the integral

$$\begin{aligned} \int_1^0 \int_0^x e^{-x^2} dy dx &= \int_0^1 e^{-x^2} y \Big|_0^x dx \\ &= \int_0^1 x e^{-x^2} dx \\ &= -\frac{1}{2} e^{-x^2} \Big|_0^1 \\ &= -\frac{1}{2} \left(\frac{1}{e} - 1 \right) \\ &= \frac{e - 1}{2e} = 0.316. \end{aligned}$$

7.3 Triple Integrals

Definition of Triple Integral

If f is continuous over a bounded solid region Q , then the **triple integral of f over Q** is defined as

$$\iiint_Q f(x, y, z) dV = \lim_{|\Delta| \rightarrow 0} \sum_{i=1}^n f(x_i, y_i, z_i) \Delta V_i$$

provided the limit exists. The **volume** of the solid region Q is given by

$$\text{Volume of } Q = \iiint_Q dV.$$

Evaluation by Iterated Integrals

Let f be continuous on a solid region Q defined by

$$a \leq x \leq b, \quad h_1(x) \leq y \leq h_2(x), \quad g_1(x, y) \leq z \leq g_2(x, y)$$

where h_1, h_2, g_1 and g_2 are continuous functions. Then

$$\iiint_Q f(x, y, z) dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{g_1(x,y)}^{g_2(x,y)} f(x, y, z) dV.$$

Example 7.3.1. Evaluate the triple iterated integral

$$\int_0^2 \int_0^x \int_0^{x+y} e^x(y+2z) dz dy dx.$$

Solution: For the first integration, hold x and y constant and integrate with respect to z .

$$\begin{aligned} \int_0^2 \int_0^x \int_0^{x+y} e^x(y+2z) dz dy dx &= \int_0^2 \int_0^x e^x(yz + z^2) \Big|_0^{x+y} dy dx \\ &= \int_0^2 \int_0^x e^x(x^2 + 3xy + 2y^2) dy dx. \end{aligned}$$

For the second integration, hold x constant and integrate with respect to y .

$$\begin{aligned} \int_0^2 \int_0^x e^x(x^2 + 3xy + 2y^2) dy dx &= \int_0^2 \left[e^x \left(x^2 y + \frac{3xy^2}{2} + \frac{2y^3}{3} \right) \right]_0^x dx \\ &= \frac{19}{6} \int_0^2 x^3 e^x dx \\ &= \frac{19}{6} \left[e^x(x^3 - 3x^2 + 6x - 6) \right]_0^2 \\ &= \frac{19}{6} \left(\frac{e^2}{3} + 1 \right). \end{aligned}$$

Example 7.3.2. If $f(x, y, z) = xy + yz$ and T consists of those points (x, y, z) in space satisfying the inequalities $-1 \leq x \leq 1$, $2 \leq y \leq 3$ and $0 \leq z \leq 1$, then

$$\begin{aligned} \iiint_T f(x, y, z) dV &= \int_{-1}^1 \int_2^3 \int_0^1 (xy + yz) dz dy dx \\ &= \int_{-1}^1 \int_2^3 \left[xyz + \frac{1}{2}yz^2 \right]_{z=0}^1 dy dx \\ &= \int_{-1}^1 \int_2^3 \left(xy + \frac{1}{2}y \right) dy dx \\ &= \int_{-1}^1 \left[\frac{1}{2}xy^2 + \frac{1}{4}y^2 \right]_{y=2}^3 dx \\ &= \int_{-1}^1 \left(\frac{5}{2}x + \frac{5}{4} \right) dx \\ &= \left[\frac{5}{4}x^2 + \frac{5}{4}x \right]_{-1}^1 = \frac{5}{2}. \end{aligned}$$

7.4 Change of Variables : Jacobians

7.4.1 Jacobians

The Jacobian is named after the German mathematician Carl Gustav Jacobi (1804-1851). For the single integral

$$\int_a^b f(x) dx$$

you can change variables by letting $x = g(u)$, so that $dx = g'(u)du$, and obtain

$$\int_a^b f(x) dx = \int_c^d f(g(u))g'(u) du$$

where $a = g(c)$ and $b = g(d)$. Note that the change of variable introduces an additional factor $g'(u)$ into the integrand. This also occurs in the case of double integrals.

$$\iint_R f(x, y) dA = \iint_S f(g(u, v), h(u, v)) \underbrace{\left[\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \right]}_{\text{Jacobian}} dudv$$

where the change of variables $x = g(u, v)$ and $y = h(u, v)$ introduces a factor called the **Jacobian** of x and y with respect to u and v .

Definition of the Jacobian

If $x = g(u, v)$ and $y = h(u, v)$, then the **Jacobian** of x and y with respect to u and v , denoted by $\frac{\partial(x, y)}{\partial(u, v)}$ is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v}.$$

In cases it is more convenient to express u and v in terms of x and y , we can first compute $\frac{\partial(u, v)}{\partial(x, y)}$ explicitly and then find the needed Jacobian $\frac{\partial(x, y)}{\partial(u, v)}$ from the formula

$$\frac{\partial(x, y)}{\partial(u, v)} \cdot \frac{\partial(u, v)}{\partial(x, y)} = 1.$$

Example 7.4.1. Find the Jacobian for the change of variables defined by

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta.$$

Solution: From the definition of a Jacobian, we obtain

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r.$$

The above example points out that the change of variables from rectangular to polar coordinates for a double integral can be written as

$$\begin{aligned} \iint_R f(x, y) dA &= \iint_S f(r \cos \theta, r \sin \theta) r dr d\theta, r > 0 \\ &= \iint_S f(r \cos \theta, r \sin \theta) \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| dr d\theta \end{aligned}$$

where S is the region in the $r\theta$ -plane that corresponds to the region R in the xy -plane. In general, a change of variables is given by a one-to-one **transformation** T from a region S in the uv -plane to a region R in the xy -plane, to be given by

$$T(u, v) = (x, y) = (g(u, v), h(u, v))$$

where g and h have continuous first partial derivatives in the region S . Note that the point (u, v) lies in S and the point (x, y) lies in R . In most cases, we are hunting for a transformation for which the region S is simpler than the region R .

Change of Variables for Double Integrals

Theorem 7.4.1. Let R and S be regions in the xy - and uv -planes that are related by the equations $x = g(u, v)$ and $y = h(u, v)$ such that each point in R is the image of a unique point in S . If f is continuous on R , g and h have continuous partial derivatives on S , and $\frac{\partial(x, y)}{\partial(u, v)}$ is non-zero on S , then

$$\iint_R f(x, y) dA = \iint_S f(g(u, v), h(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

Example 7.4.2. Let R be the region bounded by the lines

$$x - 2y = 0, \quad x - 2y = -4, \quad x + y = 4, \quad \text{and} \quad x + y = 1.$$

Evaluate the double integral

$$\iint_R 3xy dA.$$

Solution: to begin, let $u = x + y$ and $v = x - 2y$. Solving this system of equations for x and y produces $x = \frac{1}{3}(2u + v)$ and $y = \frac{1}{3}(u - v)$. The partial derivatives of x and y are

$$\frac{\partial x}{\partial u} = \frac{2}{3}, \quad \frac{\partial x}{\partial v} = \frac{1}{3}, \quad \frac{\partial y}{\partial u} = \frac{1}{3} \quad \text{and} \quad \frac{\partial y}{\partial v} = -\frac{1}{3}$$

which implies that the Jacobian is

$$\begin{aligned} \frac{\partial(x, y)}{\partial(u, v)} &= \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \\ &= \begin{vmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{vmatrix} \\ &= -\frac{2}{9} - \frac{1}{9} = -\frac{1}{3}. \end{aligned}$$

Therefore, we obtain

$$\begin{aligned} \iint_R 3xy \, dA &= \iint_S 3 \left[\frac{1}{3}(2u + v) \frac{1}{3}(u - v) \right] \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, dv \, du \\ &= \int_1^4 \int_{-4}^0 \frac{1}{9} (2u^2 - uv - v^2) \, dv \, du \\ &= \frac{1}{9} \int_1^4 \left[2u^2v - \frac{uv^2}{2} - \frac{v^3}{3} \right]_{-4}^0 \, du \\ &= \frac{1}{9} \int_1^4 \left(8u^2 + 8u - \frac{64}{3} \right) \, du \\ &= \frac{1}{9} \left[\frac{8u^3}{3} + 4u^2 - \frac{64}{3}u \right]_1^4 \\ &= \frac{164}{9}. \end{aligned}$$

Example 7.4.3. Suppose R is the plane bounded by the hyperbolas $xy = 1$, $xy = 3$ and $x^2 - y^2 = 1$, $x^2 = y^2 = 4$. Find $\iint_R (x^2 + y^2) \, dx \, dy$.

Solution: The hyperbolas bounding R are u -curves and v -curves if $u = xy$ and $v = x^2 - y^2$. We can most easily write $x^2 + y^2$ in terms of u and v by first noting that

$$4u^2 + v^2 = 4x^2y^2 + (x^2 - y^2)^2 = (x^2 + y^2)^2,$$

so that $x^2 + y^2 = \sqrt{4u^2 + v^2}$. Now

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} y & x \\ 2x & -2y \end{vmatrix} = -2(x^2 + y^2).$$

Hence we have

$$\frac{\partial(x, y)}{\partial(u, v)} = -\frac{1}{2(x^2 + y^2)} = -\frac{1}{2\sqrt{4u^2 + v^2}}.$$

Therefore

$$\iint_R (x^2 + y^2) dx dy = \int_1^4 \int_1^3 \sqrt{4u^2 + v^2} \frac{1}{2\sqrt{4u^2 + v^2}} du dv = \int_1^4 \int_1^3 \frac{1}{2} du dv = 3.$$

Example 7.4.4. Find the area of the region R bounded by the curves $xy = 1$, $xy = 3$ and $xy^{1.4} = 1$, $xy^{1.4} = 2$.

Solution: Define change of variables transformation by $u = xy$ and $v = xy^{1.4}$. Then

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} y & x \\ y^{1.4} & (1.4)xy^{0.4} \end{vmatrix} = (0.4)xy^{1.4} = (0.4)v.$$

So $\frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{\frac{\partial(u, v)}{\partial(x, y)}} = \frac{2.5}{v}$. Consequently,

$$\iint_R dx dy = \int_1^2 \int_1^3 \frac{2.5}{v} du dv = 5 \ln 2.$$

7.5 Tutorial Questions

1. Evaluate the integral.

$$(i) \int_0^x (2x - y) dy \quad (ii) \int_x^{x^2} \frac{y}{x} dy \quad (iii) \int_{e^y}^y \frac{y \ln x}{x} dx \quad (iv) \int_0^{x^3} ye^{-\frac{y}{x}} dy \quad (v) \int_0^{\cos x} y dy.$$

2. Evaluate the iterated integral.

$$(i) \int_0^1 \int_0^2 (x+y) dy dx \quad (ii) \int_0^1 \int_0^x \sqrt{1-x^2} dy dx \quad (iii) \int_0^2 \int_{3y^2-6y}^{2y-y^2} 3y dx dy \quad (iv) \int_0^{\frac{\pi}{2}} \int_0^{\sin \theta} \theta r dr d\theta.$$

3. Sketch the region R of integration and switch the order of integration.

$$(i) \int_0^4 \int_0^y f(x, y) dx dy \quad (ii) \int_0^4 \int_{\sqrt{y}}^2 f(x, y) dx dy \quad (iii) \int_{-1}^1 \int_{x^2}^1 f(x, y) dy dx.$$

4. Sketch the region R whose area is given by the iterated integral. Then switch the order of integration and show that both orders yield the same area.

$$(i) \int_0^1 \int_0^2 dy dx \quad (ii) \int_0^2 \int_{\frac{x}{2}}^1 dy dx \quad (iii) \int_0^4 \int_{\sqrt{x}}^2 dy dx \quad (iv) \int_{-2}^2 \int_0^{4-y^2} dx dy.$$

5. Evaluate the iterated integral. (Note it is necessary to switch the order of integration)

$$(i) \int_0^2 \int_x^2 x \sqrt{1+y^3} dy dx \quad (ii) \int_0^2 \int_x^2 e^{-y^2} dy dx \quad (iii) \int_0^1 \int_y^1 \sin x^2 dx dy \quad (iv) \int_0^2 \int_{y^2}^4 \sqrt{x} \sin x dx dy \\ (v) \int_0^1 \int_y^1 \frac{dx dy}{1+x^4} \quad (vi) \int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx \quad (vii) \int_0^1 \int_y^1 e^{-x^2} dx dy.$$

6. Use polar coordinates to evaluate each of the following integrals.

$$(i) \int_0^1 \int_0^{\sqrt{1-x^2}} \frac{dydx}{4-x^2-y^2} \quad (ii) \int_0^2 \int_0^{\sqrt{4-x^2}} (x^2+y^2)^{\frac{3}{2}} dydx \quad (iii) \int_0^1 \int_0^{\sqrt{1-y^2}} \sin(x^2+y^2) dx dy.$$

7. Evaluate the triple integral.

$$(i) \int_0^3 \int_0^2 \int_0^1 (x+y+z) dx dy dz \quad (ii) \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 x^2 y^2 z^2 dx dy dz \quad (iii) \int_1^4 \int_0^1 \int_0^x 2ze^{-x^2} dy dx dz$$

$$(iv) \int_1^4 \int_1^{e^2} \int_0^{\frac{1}{xz}} \ln z dy dz dx \quad (v) \int_0^{\frac{\pi}{2}} \int_0^{\frac{y}{2}} \int_0^{\frac{1}{y}} \sin y dz dx dy \quad (vi) \int_0^9 \int_0^{\frac{y}{3}} \int_0^{\sqrt{y^2-9x^2}} z dz dx dy.$$

8. Find the Jacobian $\frac{\partial(x,y)}{\partial(u,v)}$ for the indicated change of variables.

$$(i) x = -\frac{1}{2}(u-v), y = \frac{1}{2}(u+v) \quad (ii) x = au + bv, y = cu + dv \quad (iii) x = u - uv, y = uv$$

$$(iv) x = u \cos \theta - v \sin \theta, y = u \sin \theta + v \cos \theta \quad (v) x = e^u \sin v, y = e^u \cos v.$$

9. Evaluate $\iint_D \sqrt{x^2+y^2} dx dy$, where D is the region bounded by $x^2+y^2=4$ and $x^2+y^2=9$.

10. Evaluate $\iint_D (x+y)^2 dx dy$ where D is the parallelogram bounded by the lines $x+y=0$, $x+y=1$, $2x-y=0$ and $2x-y=3$.

11. Let D be the region in the first quadrant bounded by the hyperbolas $xy=1$, $xy=9$ and the lines $y=x$, $y=4x$. Evaluate $\iint_D \left(\sqrt{\frac{y}{x}} + \sqrt{xy} \right) dx dy$.

12. Using a suitable transformation, evaluate $\iint_R x(x^2-y^2)^{\frac{1}{2}} dx dy$, where R is the region bounded by $x^2-y^2=1$, $x^2-y^2=2$, $2y=x$ and $5y=x$.

13. Using polar co-ordinates, evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$. Deduce the value of $\int_0^\infty e^{-x^2} dx$.

14. By changing the order of integration or otherwise, evaluate $\int_0^a \int_0^{a-\sqrt{a^2-x^2}} \frac{xy \log(y+a)}{(y-a)^2} dy dx$.

15. Change the order of integration and evaluate the integral $\int_0^1 \int_{\sqrt{x}}^1 \frac{xy^2}{\sqrt{x^2+y^2}} dy dx$.

16. Evaluate $\iint_R \sqrt{x^2+y^2} dy dx$, where R is the region in which $x^2+y^2 \leq 4$, $x \geq 0$, $y \geq 0$.

17. By means of the substitutions $u = x+y$ and $v = y-x$, evaluate the integral $\int_0^a \int_0^a \frac{1}{\sqrt{a^2-(y-x)^2}} dx dy$.

18. By means of substitutions $x = \frac{1}{2}(u+v)$ and $y = \frac{1}{2}(u-v)$, evaluate the integral $\int_0^a \int_0^a \frac{dxdy}{\sqrt{a^2 + (y+x)^2}}$.
19. Calculate $\iint_R (x+y)^3 \cos^2(x-y) dxdy$, where R is the region bounded by the lines $x+y = \pi$, $x+y = 5\pi$, $x-y = \pi$ and $x-y = -\pi$.
20. Calculate the area of the region bounded by $xy = 4$, $xy = 8$, $xy^3 = 5$ and $xy^3 = 15$.
21. Calculate the area of the parallelogram bounded by the lines $x+y = 1$, $x+y = 2$ and $2x-3y = 2$, $2x-3y = 5$.
22. Evaluate $\iint_R (x^2 + y^2) dxdy$, where R is the region in the first quadrant bounded by the hyperbolas $x^2 - y^2 = 6$, $x^2 - y^2 = 1$, $2xy = 4$ and $2xy = 1$. (Hint: Use $u = x^2 - y^2$, $v = 2xy$)
23. Evaluate (i) $\int_0^1 \int_{\sqrt{x}}^1 e^{-\frac{x}{y^2}} dydx$ (ii) $\int_0^{\sqrt{\pi}} \int_y^{\sqrt{\pi}} \sin(x^2) dxdy$.